

100 Days of Linux - Day 060

Title: Managing Software with apt

Today's Goal

Learn how to update package lists, install, upgrade, search, inspect, and remove software using the **apt** package manager.

Why This Matters

Ubuntu and WSL users install almost all software through apt. Developers and data engineers use it to install Git, Python, Docker, htop, database clients, and many other tools.

Commands of the Day: apt

Common Syntax:

```
sudo apt update
sudo apt upgrade
sudo apt install htop
sudo apt remove htop
apt search git
apt show git
```

Common Commands:

```
update = Refresh package lists
upgrade = Install available updates
install = Install software
remove = Remove software (keep config files)
purge = Remove software and configuration
search = Find packages
show = View package details
```

Beginner Lesson

Think of apt as an app store for Ubuntu. First refresh the package list with **apt update**, then install or upgrade software. Most system-wide package operations require **sudo**.

Practice Questions

1. Refresh the package list using apt.
2. Search for the package named git and display information about it.
3. Install the tree package (or another small package if tree is already installed).
4. Remove the tree package, then reinstall it to practice both commands.
5. Display your current working directory after completing today's tasks.

Hints

Always run `sudo apt update` before installing new software.

Mini Challenge

Install htop (if needed), verify it runs, remove it, then install it again. Record each command in `apt_practice.txt`.

Common Mistakes

Running apt install before updating package lists or forgetting to use sudo for system-wide changes.

Did You Know?

APT automatically downloads software from trusted repositories and resolves package dependencies for you.

Pro Tip

Use apt search when you know part of a package name but aren't sure of the exact package to install.

Answer Key (Instructor Only)

Q1: sudo apt update

Q2: apt search git && apt show git

Q3: sudo apt install tree

Q4: sudo apt remove tree && sudo apt install tree

Q5: pwd

Homework

Install two useful utilities such as tree and neofetch (if available), explore their help pages, and remove them if you no longer need them.

Next Day Preview

Tomorrow you'll complete the Environment Variables & Package Management module with a practical assignment combining env, printenv, export, alias, unalias, and apt.